

Heavy metal concentrations in floodplain surface soils, Lahn River, Germany

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Abstract Even relatively pristine drainage basins in industrial countries would appear to have received anthropogenic inputs of heavy metals. Investigation of floodplain surface soils in the Lahn River drainage basin, west-central Germany, indicates that the Cu concentration is 1.5 times the pre-industrial level, Pb and Zn contents twice the pre-industrial level; Cd, Co, and Cr concentrations are nearly equal to background metal values. Based on contamination standards developed for the Lahn River, floodplain soils are moderately contaminated with Pb and Zn, slightly contaminated with Cu. Metal contents are uniform across the floodplain, with the exception of a peak immediately adjacent to the Lahn River. Floodplain surface soil metal contents are less in the Lahn River basin than in larger drainage systems of Germany. Although Lahn River metalliferous sediments are presently immobile, they would, if eroded, contribute to downstream heavy metal concentrations. Consequently, metal storage in smaller drainage basins such as the Lahn should be considered in predictions of future metal loads in major river systems, for aggregate small basins could serve as significant metal contributors.

Key words Floodplain metal storage · Surface soil metal contents · German fluvial systems

Introduction

Heavy metals are among the best known and most widely studied pollutants conveyed through the fluvial system. Significant heavy metal pollution likely began in the late eighteenth century or early nineteenth century, reached a peak during the mid-twentieth century, and has diminished since around 1970 (Valette-Silver 1993). In the flu-

vial system, heavy metals are typically adsorbed onto finer grained sediments (Förstner and Wittmann 1979; Salomons and Förstner 1984; Horowitz 1985). During floods these sediments, with attached metals, are spread across and deposited on the floodplain. Although high floodplain sediment storage rates are generally associated with large river systems having low gradients (Trimble 1977; Walling 1983; Leenaers and others 1988), significant storage has also been documented in small, higher gradient systems (Barnhardt 1988; James 1989). Once stored, sediments may remain in geomorphically stable drainage basins for a century or more (Meade 1982). If subsequently eroded, however, they may contribute effluents rich in heavy metals to a river (Förstner and Wittmann 1979; van der Weijden and Middelburg 1989; Samanidou and others 1991), thereby serving as an unexpected source of heavy metal pollution (Marcus 1989).

Whereas the heavy metal contents of channel bed and bank sediments and sediment in transport are well documented for many rivers in Europe (e.g., Förstner and Müller 1973, 1981; Knoblich and Sanner 1992; Runkel 1992; Brüggmann 1995), comparatively little is known about heavy metal concentrations of floodplain soils and alluvium. Several studies examine heavy metal contents of river floodplains, but usually in drainage systems that flow through areas of active or former mining (e.g., Davies and Lewin 1974; Wolfenden and Lewin 1977; Macklin and Dowsett 1989; Bradley and Cox 1986, 1990) or in regions that are highly industrialized (Leenaers and others 1988; Dehner 1994). Virtually no research has been done on floodplain metal contents in drainage systems that have little or no mining history and low levels of industrial activity (an exception is Murad and Fischer 1978). This paper provides data on heavy metal concentrations in surface soils for a 60-km reach of the Lahn River, a tributary to the Rhine River that flows through a relatively non-industrialized part of west-central Germany. The Lahn River affords an opportunity to evaluate floodplain metal concentrations in a drainage system relatively isolated from industrial activities. The study reach, because it is immediately upstream from a pronounced narrowing of the Lahn River valley, is a zone of net sediment, and potentially heavy metal, deposition. The purpose of this paper is not to determine the source of heavy metals, nor to document trends in metal concentrations vertically within deposits. Rather, it is to establish floodplain metal contents and, by looking at metal trends across the flood-

Received: 21 August 1995 · Accepted: 23 January 1996

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plain, determine whether there are metal sinks, or parts of the floodplain with relatively high metal concentrations. The results underscore that even relatively pristine drainage basins have had at least moderate anthropogenic inputs of metals. Although presently immobile, these metals would likely be eroded under a more energetic hydrological regime, especially if the regime were characterized by widespread lateral channel migration.

Study location and methodology

The reach of the Lahn River investigated herein drains an area underlain chiefly by Paleozoic shales and graywacke, and Tertiary basalt and sandstone (Knoblich and Sanner 1992). The drainage area of the Lahn River at the downstream end of the study reach is approximately 2700 km². Floodplain soils, which consist of A/AC/C profiles, are developed in silty loam sediments deposited during the last 1000 years. In valleys elsewhere in Germany, aggradation during this time has been attributed to clearing of hillslopes, which disturbed soils and increased sediment delivery to channels (Huckriede 1971; Schirmer 1983; Thiemeier 1988). In most of the study reach, this late Holocene sediment reaches a thickness of up to 2 m; in some proximal floodplain positions, however, it is overlain by an additional 1–3 m of sandy silt deposited since the late-seventeenth century (Mäkel 1969). Similar to most rivers in Germany, the Lahn River was channelized in the mid-nineteenth century, and since then has maintained virtually the same channel position. A survey of aerial photography from 1935 to 1995 revealed no lateral channel migration in the study reach during the last 60 years. Floods in the Lahn River drainage typically occur during the winter and early spring. Such events inundate most of the floodplain, but, as corroborated by field observations of flooding during the winter of 1995, they deposit little sediment on distal floodplain positions. Most sedimentation occurs immediately adjacent to the channel.

In this study, six heavy metals (Cd, Co, Cr, Cu, Pb, and Zn) that typically show evidence of anthropogenic enrichment were investigated. Concentrations in floodplain surface soils were measured at five cross-sections located between Wetzlar and Marburg (Fig. 1). To determine whether valley width has affected sediment storage (Magilligan 1985), and therefore metal contents, samples were collected from three narrow-valley (Roth, Sichertshausen, Dorlar) and two wide-valley reaches (Garbenheim, Heuchelheim). Sites were located the maximum distance possible from obvious sources of heavy metals (e.g., major roadways, industrial concentrations). At each site, samples of approximately 100 g were taken from the upper 5 cm of surface soils and placed in sterile plastic bags. The interval between samples across the floodplain ranged from 25 m to 100 m, depending on floodplain width.

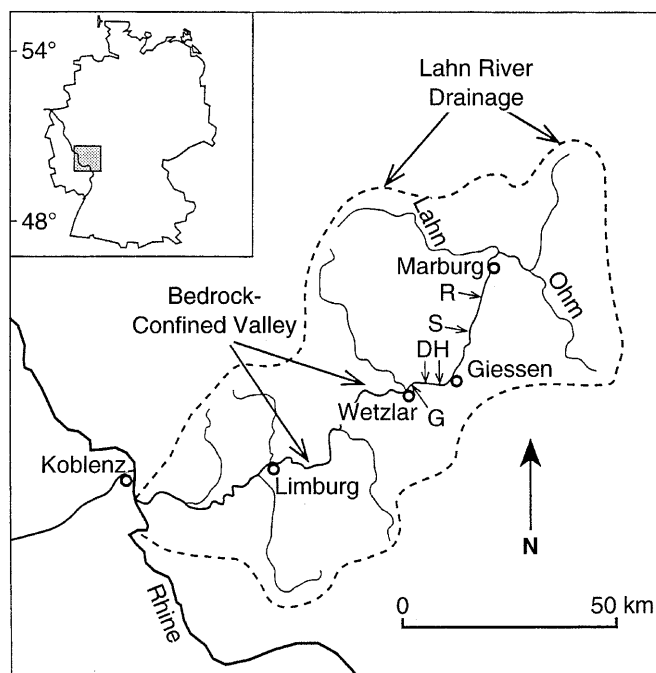


Fig. 1

Location of Lahn River study reach between Marburg and Wetzlar, Germany. Study sites are: D Dorlar; G Garbenheim; H Heuchelheim; R Roth; S Sichertshausen

In order to assess the possible anthropogenic input of metals, pre-industrial metal levels (hereafter referred to as background metal values) were calculated for the study area (see Förstner and Müller 1981; de Groot and others 1982). Commonly, background metal values are derived from pre-industrial sediments or from the extension of worldwide average metal values (e.g., Turekian and Wedepohl 1961) to a given geographical region. Although worldwide average metal values are often applied to specific drainage basins, pre-industrial sediments, because they are obtained from the same location as samples analyzed for heavy metal content, would appear to provide the most accurate values for pre-industrial metal levels of a drainage system. Using a soil auger, eight background samples were obtained from a depth of at least 1 m. Care was taken to retain only the internal part of a sample that had not come in contact with the auger barrel. Samples were stored in sterile plastic bags and transported to the laboratory.

Following the method used by many researchers (e.g., Salomons and Förstner 1980; de Groot and others 1982; Ntekim and others 1993; Mantei and Sappington 1994), heavy metal concentrations were measured for only silt-size and smaller fractions (<0.063 mm). Particles of this size typically contain the bulk of heavy metals. Samples were first sieved to remove the >2 mm fraction. Splits for particle size analysis and organic matter determination were then separated. Additional sieving removed the >0.063 mm fraction. From the remaining <0.063 mm fraction, approximately 40 g of sediment were removed

and stored in plastic bags for heavy metal analysis. Samples were first digested with perchloric, nitric, and hydrofluoric acids. The residue was then taken up in a volume of 10% hydrochloric acid. For the measurement of Cd, Co, Cr, Cu, and Zn levels, the solution was analyzed by inductively coupled plasma atomic emission spectroscopy (ICP-AES); Pb concentrations were determined by atomic absorption spectroscopy (AAS). Results, which are corrected for spectral interelement interferences, are reported in ppm.

In addition to the heavy metal analysis, particle size and organic matter analyses were completed on all samples. Particle size distribution of clay and silt was determined following the pipette method (Köhn 1929). Sand fractions were separated by wet sieving. Because sediment texture often affects metal contents (Förstner and Müller 1981; Macklin and Dowsett 1989; Mantei and Sappington 1994), trends in particle size distribution and metal contents were compared across sample sites. Likewise, organic matter content, also an important determinant of metal concentrations (Salomons and Förstner 1984; Leenaers and others 1988), was derived by loss-on-ignition at 430 °C for 24 h (Davies 1974).

Results and discussion

Tables 1 and 2 list the mean metal concentrations for the study reach as a whole and individually for each site. Background metal values are also provided. As Table 2 indicates, the Heuchelheim background metal values differ markedly from background values of the other four sites. Their effect on the mean background metal values is evident when they are excluded from calculations (Table 1). Although the high metal contents of the Heuchelheim background samples are similar to levels found in post-1800 sediments, radiocarbon ages indicate that the samples date to the pre-industrial period. The Heuchelheim background samples were collected at depths of

1.5 m and 1.75 m from a black, organic-rich peat. Subsequently, uncalibrated radiocarbon ages on the peat of 1620 ± 70 years BP (Utc-4124) at a depth of 1.5 m and 1950 ± 70 years BP (Utc-4123) at a depth of 1.75 m were obtained (A. Stobbe, Unpub. data). The ages demonstrate that the background samples were collected from sediments deposited at least 1600 years ago. Additionally, the ages indicate that the collection of samples for heavy metal background values at a depth of 1 m, as was done in this research, likely ensured that only pre-industrial sediments were sampled. The anomalously high metal contents of the Heuchelheim background samples may stem from high organic (10%) and clay contents (61%). The high clay and organic contents are suggestive of accumulation in a backwater setting, which may have served as a sink for metals being conveyed through the Lahn drainage basin. An alternative explanation for the high metal contents of the samples is that they may have a different source area than do the samples from which the other background values were measured. Because the background values from the other four sites are approximately equal, they are considered representative of pre-industrial metal concentrations in sediments of the study reach, and therefore are used in assessing the anthropogenic contribution of metals.

Comparison of the mean surface metal contents with background values indicates that floodplain soils are contaminated with Cu, Pb, and Zn, but not with Cd, Co, and Cr (Table 1). The mean level of Cu for the 50 samples is 1.5 times the background level (excluding Heuchelheim), the levels of Pb and Zn are approximately twice the background level. According to contamination standards developed for the Lahn River on the basis of the Geoaccumulation Index (Muller 1979), floodplain soils are moderately contaminated with Pb and Zn, but only lightly contaminated with Cu (Die Lahn, Ein Fließgewässerökosystem 1994). The metal levels in floodplain soils are also lower than the mean concentration for Cu (75 ppm), Pb (105 ppm), and Zn (409 ppm) measured in bed sediments (< 2 mm) of the Lahn River by Knoblich and Sanner

Table 1

Summary heavy metal values of floodplain surface soils, Lahn River study reach. Background values were calculated including and excluding Heuchelheim values. All values in ppm

	Cd	Co	Cr	Cu	Pb	Zn
All sites <i>n</i> = 50						
Mean (1 Sigma)	0.6 (0.1)	11.8 (2.1)	76.7 (8.7)	28.6 (7.3)	41.6 (8.4)	123.3 (37.7)
Range	0.5–2.0	7–18	56–100	18–53	32–68	80–278
All background (including Heuchelheim) <i>n</i> = 8						
Mean (1 Sigma)	0.5 (0.0)	12.6 (1.1)	85.0 (19.4)	24.1 (9.4)	22.7 (8.3)	84.5 (29.4)
Range	—	11–15	64–117	15–44	16–40	60–134
All background (excluding Heuchelheim) <i>n</i> = 6						
Mean (1 Sigma)	0.5 (0.0)	12.8 (1.2)	74.3 (7.0)	19 (2.7)	18.3 (2.9)	68.3 (10.2)
Range	—	11–15	64–86	15–24	16–24	60–88

Table 2

Summary heavy metal values of floodplain surface soils and background values for samples sites in Lahn River study reach. Background^a values obtained from single sample, background^b values are the mean of two samples. All values in ppm

	Cd	Co	Cr	Cu	Pb	Zn
Roth <i>n</i> = 7						
Mean (1 Sigma)	0.6 (0.2)	12.6 (1.4)	76.8 (5.2)	26.8 (3.4)	39.7 (2.5)	118.6 (28.5)
Median	0.5	13	77	26	40	112
Range	0.5–1.0	10–15	70–86	23–34	36–44	94–184
Background ^a	0.5	13	76	15	16	68
Sichertshausen <i>n</i> = 9						
Mean (1 Sigma)	0.6 (0.2)	12.9 (1.5)	76.1 (7.8)	30.1 (4.0)	39.3 (6.6)	120.4 (23.3)
Median	0.5	13	77	29	40	118
Range	0.5–1.0	10–15	61–89	23–34	32–48	92–158
Background ^a	0.5	12	70	18	16	20
Heuchelheim <i>n</i> = 10						
Mean (1 Sigma)	0.6 (0.1)	10.8 (1.9)	78.6 (9.3)	31.4 (6.1)	39.6 (7.6)	119.6 (20.5)
Median	0.5	11	79	30	40	116
Range	0.5–1.0	7–15	56–90	21–44	32–60	80–158
Background ^b	0.5	12	117	39.5	36	133
Dorlar <i>n</i> = 11						
Mean (1 Sigma)	0.5 (0.2)	11.4 (1.5)	79.8 (4.7)	29.1 (7.8)	38.2 (6.2)	116.9 (47.2)
Median	0.5	11	80	27	34	100
Range	0.5–1.0	10–15	71–88	22–52	32–54	84–260
Background ^b	0.5	12	75	18.5	17	67
Garbenheim <i>n</i> = 13						
Mean (1 Sigma)	0.7 (0.4)	11.8 (2.8)	72.9 (10.3)	27.2 (10.0)	48.8 (9.4)	136 (45.7)
Median	0.5	12	73	23	46	124
Range	0.5–2.0	7–18	58–100	18–53	34–68	88–278
Background ^a	0.5	14	75	22	22	74

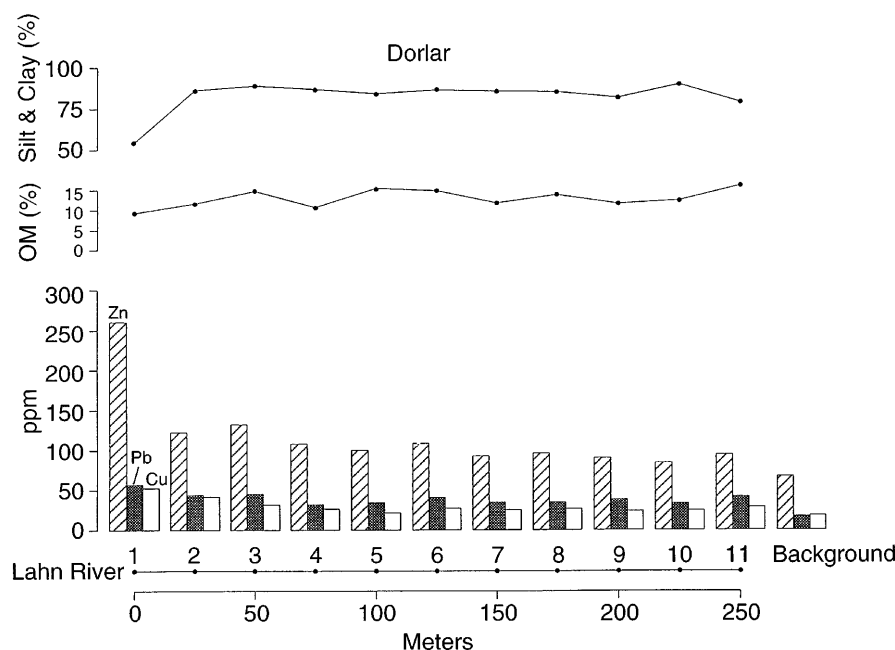
(1992). The high level of contamination in river bed sediments may result from a continuous input of metals to the channel; in contrast, the floodplain receives metals only during floods.

As indicated in Table 2, there is only minor site-to-site variation in floodplain metal contents. Apparently, valley width has little effect on metal concentrations in floodplain soils; narrow valley sites (Roth, Sichertshausen, Dorlar) have virtually the same metal contents as do wide-valley sites (Heuchelheim, Garbenheim). Only at the Garbenheim site are the values slightly higher, possibly because of the proximity of the site to a downstream steel mill, metal scrapyards, and busy four-lane highway. These may contribute metal-contaminated aerosols to the floodplain. Because no major tributary enters the Lahn River between the Dorlar and Garbenheim sites, the addition of materials from a different source area does not appear to explain the higher metal contents.

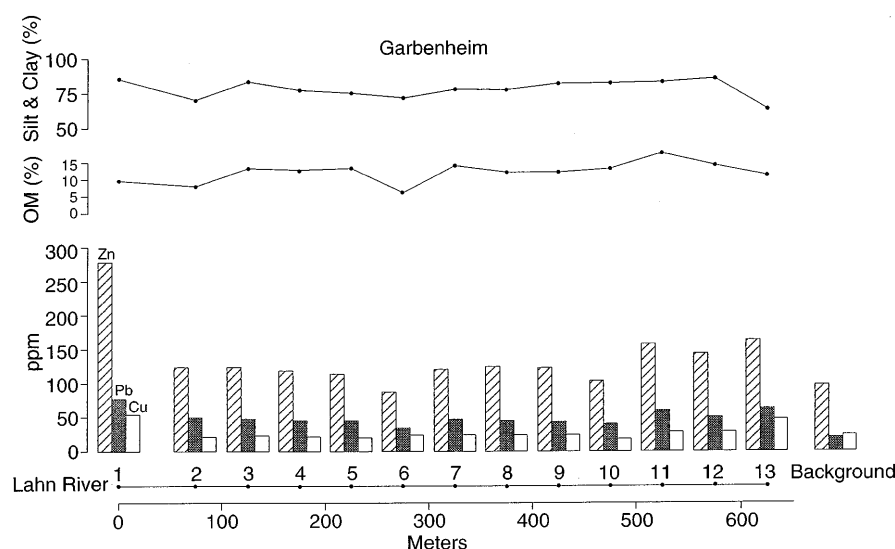
In order to look at the distribution of metals across the floodplain, transects depicting Cu, Pb, and Zn contents, percentage of silt and clay, and organic matter content for the Dorlar and Garbenheim sites were compiled (Figs. 2 and 3); the other sites have similar trends. Only the

heavy metals that show evidence of anthropogenic enrichment are portrayed. At both sites the heavy metal concentrations, especially that of Zn, are high adjacent to the active channel, but drop sharply within a distance of approximately 25 m from the channel bank. The high contents near the channel bank do not appear to result from a relatively high percentage of fine sediment or high organic matter content as sample points having higher organic matter contents and great percentages of fine sediment, but lower metal contents, are found elsewhere on the floodplain. The probable explanation for the high concentrations near the channel is that this position of the floodplain is inundated most frequently, thereby receiving regular inputs of sediments and metals. Because the Lahn River is contaminated more with Zn than with Cu or Pb, the decline in Zn concentration across the floodplain is proportionally greater with distance from the channel than is the decline in Cu and Pb. The sites not depicted in Figs. 2 and 3 show a similar decrease in Cu, Pb, and Zn concentrations with increasing distance from the river.

Away from the near-channel position of the floodplain, metal concentrations, organic matter content, and per-

**Fig. 2**

Cu, Pb, and Zn concentrations, percentage silt and clay, and organic matter content across floodplain at Dorlar site (narrow valley site). Background metal values from subsurface sediments. Sample interval across floodplain is 25 m

**Fig. 3**

Cu, Pb, and Zn concentrations, percentage silt and clay, and organic matter content across floodplain at Garbenheim site (wide valley site). Background metal values from subsurface sediments. Sample interval across floodplain is 50 m, except between samples 1 and 2, where the interval is 75 m

centages of silt and clay vary little. Excluding near-channel positions, floodplain zones having anomalously high metal concentrations, which have been noted in other drainage basins (e.g., Wolfenden and Lewin 1977; Bradley and Cox 1990; Dehner 1994), are not present in the study reach. This suggests that when flooding occurs, most sediment and metals are deposited close to the active channel; other parts of the floodplain apparently receive roughly equal, but lesser, quantities of sediments and metals. The absence of any metal sinks can probably be explained by the minimal topography of the floodplain. Topographic lows (e.g., old channels) that would favor the deposition of metalliferous fine sediment are rare in the study reach. Additionally, the Lahn River has meandered little, if at all, since channelization in the mid-1800s. Consequently, there are few abandoned channels

filled with post-nineteenth century metal-rich alluvium. The metal concentrations of Lahn River floodplain soils are much lower than concentrations measured in surface soils of major drainage basins elsewhere in Germany. For example, Cu, Pb, and Zn concentrations in the study reach are substantially lower than the mean Cu (206 ppm), Pb (123 ppm) and Zn (1190 ppm) concentrations of floodplain soils along the Elbe River, one of the most polluted rivers in Europe (Brügmann 1995). Similarly, metal concentrations in Lahn floodplain soils are less than the mean values for Cu (32 ppm), Pb (73 ppm) and Zn (232 ppm) measured in floodplain soils of the Rhine River in west-central Germany (Dehner 1994). Notwithstanding these differences, the results from this study indicate that the Lahn River has received anthropogenic inputs of heavy metals. The decline in metal contents with

distance from the active channel suggests that most metals reached the floodplain via fluvial rather than eolian deposition. Although metal concentrations are low-to-moderate, floodplain surface soils should be regarded as a potential source for future heavy metal pollution. Finally, it is worth noting that several recent studies in Germany demonstrate that surface soils often have lower heavy metal contents than do subsurface sediments (Müller 1988; Dehner 1994; Brüggmann 1995). The difference is attributed to reduction in metal releases to the environment over the last 20–30 years. If such is the case in the Lahn River drainage basin, then higher stores of metals may be present in and just below the lower A horizon of modern floodplain soils.

Conclusions

Accurate prediction of future heavy metal loads in large European river systems can emerge only when the levels of heavy metals stored in major tributary valleys are known. In aggregate, tributaries have the potential to serve as important future sources of heavy metal pollution. To this end, heavy metal contents in floodplain soils were measured for a 60-km reach of the Lahn River, a major tributary to the Rhine River. Samples were found to contain low-to-moderate levels of heavy metals. In comparison to the metal contents of pre-industrial sediments collected from the subsurface, surface soils at five sites are slightly contaminated with Cu and moderately contaminated with Pb and Zn; concentrations of Cd, Co, and Cr in surface soils are nearly identical to concentrations in pre-industrial sediments. Virtually no site-to-site variation in metal concentrations was discernible, suggesting that valley width has had no effect on the concentration of metals in surface soils. At each site, floodplain metal concentrations are highest immediately adjacent to the active channel. Metal contents drop sharply with increasing distance from the river and are essentially uniform across the floodplain. Metal sinks, or areas of the floodplain with anomalously high metal concentrations, were not identified in the study reach. Additionally, no clear relationship among particle size distribution, organic matter content, and metal concentrations could be established.

In comparison with other rivers in Germany (e.g., Rhine, Elbe), the level of metal contamination in the Lahn River basin is relatively low. Nevertheless, this research demonstrates that drainage basins distant from major industrial concentrations have received inputs of anthropogenic heavy metals. Because the high levels of contamination are at and immediately adjacent to the Lahn River, the bulk of these metalliferous sediments would appear to be fluvial rather than eolian deposits. Although presently immobile under the channelized hydrological condition of the river, these metals could be eroded if the hydrologic regime became more energetic, especially if it were characterized by lateral channel migration. Should this

occur, the metals would likely be transported through the narrow downstream reach of the Lahn River, where stream power is proportionally higher, potentially reaching the Rhine River.

Acknowledgements This study was completed while I was a Research Fellow of the Alexander von Humboldt Foundation, in residence at the Geographisches Institut, Justus-Liebig-Universität, Giessen, Germany. I thank Professor Dr. W. Haffner and other members of the Geographisches Institut for providing field and logistical support during my year in Giessen. The help of M. Schick, who assisted with laboratory work, and B. Goecke and K. Wegner, who produced the graphics, is gratefully acknowledged. Funding for the heavy metal analysis was provided by a research grant from the Alexander von Humboldt Foundation.

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